

**School of Computing
Department of Software Engineering**

Course Syllabus

SE202 - Mobile Programming

**Fall Semester
2026**

Course Title	Mobile Programming		
Course Code	SE202		
Credits / ECTS	6	Total Hours	180
Contact hours per week	Lecture / Lab		2 / 2
Pre-requisite(s)	CS201		
Semester	Fall	Academic Year	2026
Course leader	Muxammadali Hayotov		
Email	m.khayotov@newuu.uz		
Office Number	218		
Office Hours			
Textbook	Payne, L. Dart Apprentice. 3rd ed. Kodeco, 2023 Lau, L. Mobile Deep Learning with TFLite. 1st ed. O'Reilly Media, 2023. Huyen, C. Designing ML Systems (with GenAI SDK Docs). 1st ed. O'Reilly Media, 2022.		
Reference Books	Napoletano, A. Mastering Dart & Flutter. 1st ed. Packt Publishing, 2024. Napoletano, A. Mastering Dart & Flutter (with Flutter Impeller Docs). 1st ed. Packt Publishing, 2024		

Course Description (as in the catalog):

This course introduces students to the principles and techniques of mobile application development, including user interface design, mobile programming languages (such as Java or Swift), and mobile development frameworks (such as Android Studio or Xcode). Students will learn how to design and implement user-friendly and responsive mobile applications, incorporating features such as location services, sensors, and data storage. Through hands-on projects and assignments, students will gain practical experience in building mobile applications, testing and debugging, and deploying apps to app stores. By the end of the course, students will have the skills and knowledge necessary to develop mobile applications independently.

Course Instructors:

Position	Name	Email	Office	Office hours
Senior Lecturer / Katta o'qituvchi	Muxammadali Hayotov	m.khayotov@newuu.uz	218	

Attendance policy:

Attendance is compulsory. We emphasize the importance of the discussions during the instructed hours. A student missing 25% (32 hours or more) of all instructed hours will be forced to withdraw the course (in accordance with the University Academic Regulations).

Course Learning Outcomes (CLOs):

By the end of successful completion of this course, the student will be able to:

No.	Description
CLO1	Design and develop mobile applications with intuitive user interfaces.
CLO2	Implement mobile app functionality using native and cross-platform frameworks.
CLO3	Integrate cloud services and APIs into mobile applications.
CLO4	Optimize mobile applications for performance and security.
CLO5	Deploy and publish mobile applications to app stores.

Contribution level of Course Learning Outcomes(CLOs) to Student/Program Outcomes

Software Engineering

	S01	S02	S03	S04	S05	S06	S07
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	-	-	-	-	-
CLO2	-	High	-	-	-	Medium	-
CLO3	Medium	High	-	-	-	-	-
CLO4	High	-	-	-	-	Medium	-
CLO5	-	High	-	-	Medium	-	-

Cyber Security

	S01	S02	S03	S04	S05	S06	S07
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	-	-	-	-	-
CLO2	-	High	-	-	-	Medium	-
CLO3	Medium	High	-	-	-	-	-
CLO4	High	-	-	-	-	Medium	-
CLO5	-	High	-	-	Medium	-	-

AI & Robotics

	S01	S02	S03	S04	S05	S06	S07
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	-	-	-	-	-
CLO2	-	High	-	-	-	Medium	-
CLO3	Medium	High	-	-	-	-	-
CLO4	High	-	-	-	-	Medium	-
CLO5	-	High	-	-	Medium	-	-

Weekly Distribution of Course Topics/Contents

Week	Class Type	Topic	Required Reading	Outcome
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1	Lecture / Tutorial	Modern Dart 3 Fundamentals: Sound null safety, Records, Patterns, Class Modifiers (sealed, final), Switch expressions, Dot shorthands, and Primary constructors. Lab: Configure Dart MCP Server in IDE. Refactor a legacy Dart console app to use records, pattern matching, and sealed class modifiers.	• Dart Apprentice: Fundamentals, Ch. 1-6 • Dart Apprentice: Beyond the Basics, Ch. 3-9 • Dart 3 Language Specs (dart.dev)	CLO1, CLO2
2	Lecture / Tutorial	Declarative UI & Composition Element Tree vs. RenderObjects, Material 3 design tokens, dynamic theme switching, and layout composition (Flex, Wrap, LayoutBuilder). Lab: Recreate a complex multi-screen dashboard using adaptive Material 3 tokens for mobile and desktop viewports.	• Mastering Dart & Flutter, Ch. 2 & 3 • Flutter Docs: "Understanding Constraints"	CLO2
3	Lecture / Tutorial	Impeller Engine & WebAssembly (Wasm) Impeller runtime shader pre-compilation vs. Skia; pipeline architectures; Dart-to-Wasm compilation for web. Lab: Profile frame rates (60/120 FPS), memory footprints, and rendering timelines using Flutter DevTools under high-frequency animations.	• Flutter Engine Architecture Docs: "Impeller Engine" • Dart Web Specs: "Compiling Dart to Wasm"	CLO2
4	Lecture / Tutorial	Declarative Routing & Deep Linking Modern routing with go_router (Navigator 2.0). Deep-linking, shell routes for bottom tabs, nested navigation stacks, route guards. Lab: Implement a multi-tab application featuring dynamic URL path parameters, authentication guards, and deep-link handling (https://).	• Mastering Dart & Flutter, Ch. 5 • go_router Package Documentation (pub.dev)	CLO1
5	Lecture / Tutorial	Production State Management: Riverpod Declarative state reactivity with Riverpod 2.x/3.x: Providers, Notifiers, AsyncValue, and compile-time code generation. Lab: Construct a real-time data ticker UI that handles loading, error, and data states using Riverpod AsyncValue pattern matching.	• Riverpod Official Documentation (riverpod.dev) • Mastering Dart & Flutter, Ch. 6	CLO1
6	Lecture / Tutorial	Enterprise Architecture: Bloc/Cubit Event-driven state management with flutter_bloc. Stream-based state transformations, functional state isolation, clean architecture layers. Lab: Build a stock monitoring dashboard utilizing WebSockets mapped through a Bloc/Cubit event pipeline.	• Bloc Library Specs & Architecture Guide (bloclibrary.dev) • Mastering Dart & Flutter, Ch. 7	CLO1, CLO2

7	Lecture / Tutorial	Midterm Review & Capstone Design Proposals Comprehensive review of Weeks 1-6 (Dart 3, UI, Impeller, Routing, Riverpod/Bloc). Architecture design patterns for AI-driven apps. Lab / Presentation: Students present their Capstone Project Design Proposals (architecture diagrams, tech stack choices, wireframes, and proposed AI integrations) for peer/instructor approval.	• Review all assigned materials from Weeks 1-6.	CLO1, CLO2, CLO3
8	Exam	Midterm Exam		CLO1, CLO2, CLO3
9	Lecture / Tutorial	Edge AI & Computer Vision (TFLite & ML Kit) On-device zero-latency ML (tflite_flutter), tensor shape processing, background Isolates, and native vision pipelines (OCR, Face/Barcode detection via google_mlkit). Lab: Build a camera app that classifies images via an on-device TFLite model in an Isolate and extracts structured OCR receipt data without dropping frames.	• Mobile Deep Learning with TFLite, Ch. 1, 2, & 4 • Practical ML for Computer Vision, Ch. 8 • Dart Apprentice: Beyond the Basics, Ch. 14	CLO2, CLO3
10	Lecture / Tutorial	Cloud AI Services & Streaming Protocols Integrating Multimodal LLMs (OpenAI, Gemini API, Claude). Server-Sent Events (SSE) for continuous token streaming in Dart. Lab: Develop a streaming multimodal voice and text assistant that renders AI responses token-by-token with automatic scroll positioning.	• Designing Machine Learning Systems, Ch. 9 • Dart Apprentice: Beyond the Basics, Ch. 12 & 13 • Google GenAI SDK Specs (google_generative_ai)	CLO1, CLO2
11	Lecture / Tutorial	Structured Outputs & Agentic Function Calling Enforcing structured JSON schemas from LLM outputs. Function Calling / Tool Use inside Flutter to trigger client-side logic via AI. Lab: Create a "Smart Meal Planner" app where an LLM executes a tool call to auto-populate a structured local database calendar and shopping list.	• Designing Machine Learning Systems, Ch. 10 • OpenAI / Gemini Function Calling API Docs	CLO3
12	Lecture / Tutorial	Generative UI (GenUI) Architectures Dynamically rendering native Flutter UI components based on runtime AI agent responses using Flutter GenUI / A2UI protocols. Lab: Build an adaptive Travel Itinerary app that synthesizes interactive map cards, timeline sliders, and expense forms on the fly based on user prompts.	• Flutter GenUI SDK & A2UI Protocol Specs • Designing Machine Learning Systems, Ch. 7	CLO3

13	Lecture / Tutorial	Full-Stack Dart & Backend Security Backend-for-Frontend (BFF) architecture. Securing API keys via Dart serverless backends (Dart Frog / Firebase Cloud Functions). Native Secure Storage. Lab: Deploy a Dart Frog proxy server to route multimodal AI requests securely, keeping secret tokens completely isolated from client binaries.	• Mastering Dart & Flutter, Ch. 10 • Dart Frog Documentation (dartfrog.pw)	CLO4, CLO5
14	Lecture / Tutorial	Testing Strategy, CI/CD & Deployment Unit/Widget/Integration testing pyramid. AI-assisted QA & accessibility (a11y). Automated pipelines (GitHub Actions, Fastlane), code obfuscation (--obfuscate), signing. Lab: Construct a GitHub Action pipeline that runs test suites (>80% coverage) and produces signed, obfuscated Android (.aab) and iOS (.ipa) release builds.	• Mastering Dart & Flutter, Ch. 11 & 12 • Flutter Docs: "Testing Flutter Apps" & "Obfuscating Code"	CLO1, CLO2, CLO3, CLO4, CLO5
15	Lecture / Tutorial	Capstone Project Showcase & Code Reviews Peer and instructor evaluation of final production-grade Flutter + AI applications. Presentation: Live demonstration of the final capstone app including clean architecture, on-device ML, cloud GenAI integration, and an automated CI/CD pipeline.	• Final Capstone Documentation & Codebase Submission	CLO1, CLO2, CLO3, CLO4, CLO5
16	Exam	Final Exam		CLO1, CLO2, CLO3, CLO4, CLO5

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Midterm Project	Week 7	20
Midterm Exam	Week 8	20
Final Project	Week 15	20
Final Exam	Week 16	20
Lab/Assignments	Week 1-6; Week 9-14	20
Total		100

Assessment of Course Learning Outcomes:

CLO	Meaning	Evaluation Tool(s) & Weights (tentative)
CLO1	Design and develop mobile applications with intuitive user interfaces.	Lab/Assignments (40%) Midterm Project (30%) Midterm Exam (30%)
CLO2	Implement mobile app functionality using native and cross-platform frameworks.	Lab/Assignments (40%) Midterm Project (30%) Midterm Exam (30%)
CLO3	Integrate cloud services and APIs into mobile applications.	Lab/Assignments (10%) Midterm Project (50%) Midterm Exam (40%)
CLO4	Optimize mobile applications for performance and security.	Lab/Assignments (40%) Final Project (30%) Final Exam (30%)
CLO5	Deploy and publish mobile applications to app stores.	Lab/Assignments (10%) Final Project (50%) Final Exam (40%)

Students' Workload:

Students' workload is distributed as follows:

	Activity	Hrs.	%
Assessment methods and criteria	In-Class Time	56	31 %
	Independent Study	55	31 %
	Assignments	51	28 %
	Examinations	18	10 %
	Group Discussions	0	0 %
	Fieldwork and Practicum	0	0 %
	Workshops and Seminars	0	0 %
	Individual Development	0	0 %
Total		180	100%

Approval			
Course leader - Senior Lecturer			
Name	Muxammadali Hayotov		
Signature		Date:	2026-09-08
Department head - Professor			
Name	Irfanud Din		
Signature		Date:	2026-09-10
Academic Administration - Head of Department			
Name	Komiljon Yusufjonov		
Signature	—	Date:	—
Department of Quality Assurance and Accreditation - Head of Department			
Name	Aliya Kaimoldinova		
Signature	—	Date:	—